Hazard Analysis
Software Engineering

Team 2, Hindsight
Mohammad Bilal
Ethan McMehen
Jackson Beach
Jake Finlay
Ahsan Muzammil

Table 1: Revision History

Date	Developer(s)	Change
2026/10/13	Ahsan Muzammil	Completed section 4 (CA), section 5 and 6 (3 each), section 8 (only my AI uses), and Reflection
2026/10/14	Jake Finlay	Completed Section 3: System Boundaries and Components, added 3 entries to Section 5: Failure Mode and Effect Analysis, added 5 saftey requirements to Section 6: Safety and Security Requirements, added my AI disclosure and reflection
2026/10/15	Ethan McMehen	Contributed to Section 5, Section 6, Appendices, and Added Reflection
2026/10/15	Mohammad Bilal	Completed Sections 1 and 2 and the Reflection; contributed to Sections 5 and 6 and the Appendices
2026/10/17	Jackson Beach	Contributed to Sections 5 and 6; Completed Section 7; Contributed my AI usage to Section 8; Completed individual reflection
2026/10/18	Jake Finlay	Updated the SRS links in Section 6 to match the revised SRS requirement numbering and traceability

Contents

1 Introduction **1**

2 Scope and Purpose of Hazard Analysis **1**

3 System Boundaries and Components **1**
- 3.1 System Boundary . . . 1
- 3.2 Components . . . 1

4 Critical Assumptions **2**

5 Failure Mode and Effect Analysis **3**
- 5.1 Method . . . 3
- 5.2 FMEA Table . . . 3

6 Safety and Security Requirements **12**

7 Roadmap **18**

8 Appendix — Generative AI Use Disclosure **23**
- 8.1 AI Tools Used . . . 23
- 8.2 How and Where Used . . . 23
- 8.3 Verification of AI-Generated Content . . . 23

9 Appendix — Reflection **25**

1 Introduction

This document presents the hazard analysis for Software Engineering, a mobile game designed to prepare children aged 3–7 for paediatric ophthalmology appointments. It identifies potential system failures, unsafe behaviours, and other conditions that could harm users or stakeholders, and derives the safety and security requirements that eliminate or reduce those hazards.

A hazard is a system condition that, in combination with conditions in its environment, has the potential to cause a loss [1]. A loss refers to harm or damage to a stakeholder. For Software Engineering, potential losses include physical or psychological harm to children, breaches of user privacy or data security, the use of inaccurate information, and failure to prepare children effectively for their eye appointments.

2 Scope and Purpose of Hazard Analysis

The purpose of this hazard analysis is to evaluate risks associated with the development and use of Software Engineering, assess their potential consequences, and identify measures to prevent or reduce harm. The analysis considers both technical failures and risks arising from the application's interactions with young users.

The potential losses considered include:

- **Psychological harm:** Distress or confusion caused by age-inappropriate content, misleading information, or unclear instructions.
- **Privacy and security breaches:** Unauthorized disclosure of personal information or sensitive user data resulting from unnecessary data collection or insecure handling.
- **Ineffective appointment preparation:** Reduced preparedness caused by inaccurate information, misleading representations of ophthalmological procedures, or inadequate communication of relevant information.
- **Loss of functionality:** Inability to access or use the application as intended due to software defects, crashes, or other failures.
- **Loss of user trust:** Reduced confidence in the application resulting from unpredictable behaviour, unreliable performance, or unmet user expectations.

The identified hazards and their potential consequences will guide the selection of mitigation measures and the formulation of corresponding safety and security requirements.

3 System Boundaries and Components

3.1 System Boundary

The system boundary contains the game and the device it is played on.

The game includes everything in the release build. This includes the code, content, assets and any third-party software packaged with it, along with the data it saves on the device.

The device includes the hardware and operating system that run the game. The team did not make the devices, but they are inside the boundary because the game must run correctly on the device.

The boundary ends at the device. The game runs offline after installation and has no accounts or servers, so everything the system does happens on the device the child is playing on.

3.2 Components

The system contains the following components:

1. **The Device:** A phone or tablet running iOS or Android including its operating system, touchscreen, speakers or headphones, and storage.
2. **The Game:** An application installed on the device. The Game can be split into multiple subcomponents:
   - **Gameplay:** The levels, the interactions that respond to the child's touch input, and scoring.
   - **Educational Content:** The facts about each eye procedure and the story, characters and illustrations that present them.
   - **Quizzes:** The questions that check what the child has learned, and the feedback given for right and wrong answers.
   - **Progress and Rewards:** Unlocked levels, stars, collectibles, cosmetic rewards and player profiles.
   - **Audio and Visuals:** Graphics, animations, effects, music, sound effects and narration.
   - **Save Data:** The progress, rewards and settings stored on the device.
   - **Third-party Software and Assets:** The game engine, libraries, art, fonts and sounds included in the game.

4 Critical Assumptions

The following assumptions establish the expected conditions under which Software Engineering will operate. Potential failures involving these conditions will still be considered during the hazard analysis.

1. Assume that users have access to a compatible smartphone or tablet and possess, or can receive assistance with, the basic skills required to interact with a mobile application.

   Justification: The application is intended for use on consumer mobile devices. Since the primary users are young children, their familiarity with mobile devices and ability to interact independently may vary.

2. Assume that children will primarily use the application at home or in a clinical environment before their ophthalmology appointments, where the level of caregiver assistance may vary.

   Justification: The application is intended to support preparation before clinical examinations. The environment and availability of caregivers may influence how children interact with the application and whether they can complete activities independently.

3. Assume that users will operate the application on consumer mobile devices that meet the minimum hardware and operating system requirements specified by the project.

   Justification: The application is intended to operate on both iOS and Android devices with varying hardware capabilities. Device related limitations or failures may affect the availability and performance of the application.

4. Assume that internet connectivity may be unavailable or unreliable during normal application use.

   Justification: Families may use the application in environments with limited network connectivity, including clinical waiting areas. The application should therefore not depend on continuous internet access for its essential preparation activities.

5. Assume that the ophthalmology procedures children experience during appointments may vary depending on the patient, clinician, and clinical setting.

   Justification: The application prepares children for common ophthalmology procedures rather than a specific standardized appointment. Variations in clinical practices may affect how closely the simulated

experience reflects the child's actual examination. (The application cannot represent every possible variation of a clinical examination)

6. Assume that qualified clinical professionals will be available to provide feedback and review the educational content developed for the application.

   Justification: The development team relies on clinical expertise to ensure that the educational content appropriately represents ophthalmology procedures. The availability and timing of clinical feedback may affect the accuracy and readiness of the content.

5 Failure Mode and Effect Analysis

5.1 Method

The hazards are identified with a Failure Mode and Effects Analysis (FMEA). FMEA works bottom-up from the components of Section 3: for each component, the team asked how it could fail and what that failure would do to the child. This suits the game, because most of its hazards come from a single failure, such as one piece of wrong content or one unsafe effect, rather than from complex interactions between components. Failures involving the conditions in the critical assumptions of Section 4 are still analysed.

The game is meant to be finished in full before the child's visit, since families do not always know which procedures will be done. A failure that stops the child from finishing the game therefore stops them from learning about every procedure that comes after it.

Table 2 is divided into one section per component of Section 3. Each row records:

- **Component:** the main component of Section 3 and the subcomponent that fails.
- **Failure mode:** how the failure would be observed.
- **Effects of failure:** the consequence for the child or their family.
- **Causes of failure:** the reasons the failure could occur.
- **Recommended action:** what will remove the hazard or reduce its likelihood or effect.
- **SR:** the safety and security requirements of Section 6 that the actions lead to.
- **Ref:** the hazard identifier, Hc-n, where c is the main component's number in Section 3 and n is the hazard's position within that component.

5.2 FMEA Table

Table 2 is printed in landscape orientation.

Table 2: FMEA table

Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
The Device	Device becomes excessively hot during use	The child may become uncomfortable or stop using the application; in severe cases, prolonged contact with a hot device could cause minor injury	• Device has insufficient cooling • Application places excessive load on the device • Application continues performing unnecessary processing while idle	• Avoid unnecessary continuous processing • Stop resource-intensive activities when they are no longer required • Test the application on supported devices under extended use conditions	SR-1	H1-1
	Game experiences poor performance, including slow response times or dropped frames	Delayed or inconsistent gameplay may make the application difficult to use, causing frustration or preventing the child from following preparation activities effectively	• Application requires excessive processing • Insufficient memory or processing resources • Inefficient rendering or animation • Performance degrades during extended use	• Maintain responsive gameplay and smooth visual transitions • Optimize processing, rendering, and animations • Test gameplay performance on supported devices • Verify performance during extended gameplay	SR-2	H1-2
The Game: Gameplay	Repeated or accidental touch inputs cause the game to skip important preparation information	The child may miss explanations about an ophthalmology procedure, resulting in confusion or increased anxiety during the actual examination	• Multiple touch inputs are processed during a screen transition • Repeated tapping advances through several instructions unintentionally • Gameplay does not distinguish between a deliberate action and repeated inputs	• Process only one navigation action per screen transition • Temporarily disable repeated inputs while transitioning between screens • Ensure that essential procedure information cannot be skipped accidentally	SR-3	H2-1

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
	Gameplay becomes stuck or enters an unrecoverable state after an unexpected or invalid player action	The child may be unable to continue the preparation activity and may become frustrated or confused	• Invalid or unexpected input is not handled • A gameplay state does not have a valid exit • An incomplete action leaves the game in an unusable state • Error conditions are not tested during gameplay	• Provide a valid recovery path from every gameplay state • Handle unexpected or invalid inputs without preventing continued play • Allow the child to safely restart the current activity without losing completed preparation • Test gameplay recovery and interruption scenarios	SR-4	H2-2
	The child cannot understand an instruction or perform the interaction to continue	The child may become frustrated, leave the activity, or miss information though the game is running correctly	• Instructions heavily depend on reading or use unfamiliar words • Touch targets are too tiny or gestures require too much accuracy • Interactions are not tested with the intended ages and abilities	• Use short, familiar wording and narration with visual demonstrations • Make controls large and interactions forgiving • Test whether children across the target ages and abilities can follow the instructions and use the controls	SR-24	H2-3
	A preparation level or the content it needs is unavailable without an internet connection	The child cannot complete preparation at home or in a waiting room where network connections could be poor/unavailable	• Required assets are missing from the installation package • A level downloads content or waits for an external service • Offline testing uses content downloaded during an earlier session	• Include every level and its required assets in the installation • Remove network dependencies from preparation activities • Test every level offline from a fresh installation, before any content has been downloaded during play	SR-26	H2-4
The Game: Educational Content	Procedure shown inaccurately (wrong steps, order or equipment)	Child's expectations do not match the visit, causing surprise, anxiety and reduced cooperation during the examination	• No team member has clinical training • Procedures differ between patients, clinicians and clinics • Content changed after it was reviewed	• Have a paediatric eye-care clinician approve all procedure content before release • Tell the child their doctor may do things a little differently • Release only the content version that was approved	SR-5, SR-6, SR-7	H2-5

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
	Unpleasant sensations left out or played down ("it won't hurt")	Child is surprised by stinging drops, a bright light or blurry vision, and loses trust in their caregivers, clinicians and the game	• Writers try to reassure the child • Sensation information is easy to skip during fast-paced play	• Describe each sensation honestly before the child sees that part of the procedure • Never promise that something will not hurt or be uncomfortable • Do not let the sensation information be skipped	SR-8, SR-9	H2-6
	Frightening content (needles, crying, looming equipment, threatening imagery, or scary story moments)	Increased fear of the appointment; the child may resist or refuse the examination	• Realistic depiction chosen for accuracy • Danger or drama added to the story to make the game exciting • Art and imagery used without review	• Have a child-life specialist or child development expert review all images, language, and story • Keep danger and drama out of the parts of the game that show procedures	SR-9	H2-7
	The game does not clearly communicate that its content is educational rather than medical advice	Caregivers may mistake simulated procedures or game feedback for clinical guidance and rely on the application instead of consulting a clinician	• The application presents realistic ophthalmology procedures • Educational content is presented without a clear explanation of its limitations • Caregivers are not informed of the application's intended purpose	• Display an educational-use notice during initial setup • Require caregiver acknowledgment before accessing the application • Make the notice available in the caregiver procedure guide	SR-10	H2-8

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
	Character library does not represent the range of children using the game	The child experiences apprehension that they don't belong which can translate to the clinical setting	• Characters do not represent dimensions of diversity: skin tone, hair texture, accessibility aids, gender expression, etc. they see in their world • Cosmetic rewards do not include cosmetic features they see frequently such as cultural or religious attire from their world	• Provide a library of characters that is inclusive in the dimensions of diversity (i.e. skin tone, hair texture, etc.) • Ensure cosmetic rewards include cosmetics from a range of cultures and religions	SR-15	H2-9
The Game: Quizzes	Feedback function fails to reveal the correct answer after an incorrect selection.	The child experiences increased anxiety at the real clinic because they did not select and learn the correct answer for a step during a test	• An invalid answer key is designed • No answer reveal procedure is designed • Answer reveal times out before the child can see or understand it	• Show the correct answer after an attempt and provide encouragement despite the result • Show the correct answer immediately, have the child select it, then offer a fresh retry with encouragement • Ensure that every quiz question has at least one correct answer	SR-16, SR-17	H2-10
The Game: Progress and Rewards	Preparation levels become inaccessible until earlier levels are completed or sufficient points are earned	The child cannot access preparation for a procedure that may be required unexpectedly during an appointment, resulting in unfamiliarity and possible anxiety	• Journey progression is implemented with mandatory level unlocking • Access to procedure games depends on completion status or accumulated points	• Make all procedure preparation levels accessible from the beginning • Use points and rewards for motivation rather than restricting access to educational games • Verify that every level can be opened regardless of previous completion	SR-11	H2-11

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
	Incorrect answers or skipped exercises trigger negative feedback or penalties	The child experiences frustration or increased anxiety and may become reluctant to continue preparing for ophthalmology examinations	• Game rewards are awarded based on answer accuracy rather than activity completion • Skipped exercises are treated as failures • Gameplay requires correct answers before allowing progression	• Provide encouraging feedback for incorrect responses and skipped exercises • Allow children to skip exercises without losing previously earned rewards • Award level completion based on reaching the ending rather than answering correctly	SR-12	H2-12
The Game: Audio and Visuals	Audio, animations, or visual effects continue playing after the child exits a preparation level	The child may become confused or distressed by sounds or visuals from an activity they have already left	• Audio playback is not stopped when the level is exited • Ongoing animations or visual effects are not terminated during screen transitions • The exit function changes screens without stopping active game components	• Stop all level-specific audio, animations, and visual effects upon exit • Ensure that no level-specific content continues after returning to the journey map • Test exiting during active audio playback and animations	SR-13	H2-13
	Narration is not delivered during gameplay	The child feel apprehensive at the clinic because the child could not interpret the current instruction and as a pre-reader does not progress.	• Device volume is muted or set to zero by parent or child • Narration setting is turned off • Incoming call or notification interrupts the audio	• Show icon-based prompts automatically when narration turned off • Show a parent/guardian directed prompt to restore the audio • Pause on interruptions and continue at the same point	SR-18	H2-14
	Visuals in gameplay are disabled or unrendered	The child experiences incomplete preparation because narration could not convey spatial reasoning and/or the child did not progress to next step of the game	• Parent or guardian disables visual gameplay settings • Child has low visual impairment and cannot perceive the gameplay cues	• Keep visual gameplay cues and icons always active during gameplay • Provide an accesibility mode with audio description for parent selection for visually impaired children	SR-19	H2-15

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
	Sudden loud audio, excessive volume, or rapidly flashing visual effects cause discomfort or distress	The child may experience sensory discomfort, headaches, startle responses, or difficulty continuing to use the application	• Audio is played at an unnecessarily high volume • Sound effects begin suddenly without a gradual transition • Visual effects change or flash rapidly • Sensory effects are not tested on supported devices	• Keep audio at a comfortable volume and avoid unnecessarily loud sound effects • Avoid rapidly flashing or excessively repetitive visual effects • Test audio and visual effects during normal gameplay • Allow non-essential audio to be muted or reduced	SR-14	H2-16
	Narration describes a different procedure step from the animation being shown	The child may learn the wrong sequence or misunderstand what a piece of equipment does, leaving them confused or surprised at the appointment	• Narration is paired with the wrong animation • Audio and animation start or move forward at different times • Audio and animation resume at different points after an interruption	• Check that each narration matches the step being shown • Keep narration and animation synchronized during playback • Test that both resume together after interruptions on iOS and Android devices	SR-25	H2-17
The Game: Save Data	Saved progress becomes corrupted or is reset	The child loses completed levels, rewards, or settings and may become frustrated or believe that previously completed preparation has been lost	• Application terminates while saving • Invalid save data is written • Device storage becomes unavailable • Changes to the save-data format are not handled correctly	• Validate save data before loading it • Handle interrupted or invalid saves without crashing • Preserve valid progress when a save operation fails • Test save and load behaviour under interrupted conditions	SR-20	H2-18

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
	Save data contains unnecessary personal information or is accessible to other applications	Information about the child could be exposed, creating a privacy or security risk	• Personal information is stored despite not being required for gameplay • Save files are stored without appropriate device protections • Debug information is included in release save data	• Store only information required for offline gameplay • Do not store names, contact information, or other unnecessary personal information • Protect stored game data using the platform's application storage mechanisms	SR-21	H2-19
The Game: Third-party Software and Assets	A third-party library or engine component causes the application to crash or behave incorrectly	The child may lose progress, be unable to complete a preparation activity, or receive incomplete information about an ophthalmology procedure	• Third-party software contains defects • Incompatible library versions • Unexpected interaction between dependencies • Dependency is updated without sufficient testing	• Test third-party components as part of the release build • Pin and review dependency versions • Test critical gameplay and preparation flows after dependency changes • Remove unused third-party dependencies	SR-22	H2-20
	Third-party asset contains inappropriate or misleading content	The child may be exposed to frightening, inappropriate, or inaccurate information that contradicts the intended preparation experience	• Asset selected without sufficient review • Asset is modified or replaced after review • External content does not match the application's educational requirements	• Review all third-party assets before release • Include third-party assets in the same clinical and child-life review process as first-party content • Release only reviewed versions of assets	SR-23	H2-21

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Component	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref
	An included library sends data from the device	Usage data or device information may be disclosed without the family's knowledge, breaking the game's privacy protections	• Analytics/diagnostic uploads are enabled in a library • A dependency update enables a network service • Testing checks offline play but not network activity when connected	• Remove network services the game does not need • Check dependency settings before release and after updates • Verify that the release build makes no requests while connected, including after reconnecting	SR-21, SR-22	H2-22

6 Safety and Security Requirements

The following requirements were identified through the Failure Mode and Effects Analysis of Section 5. They are stated at a general level here, and measurable values such as thresholds and limits are left unspecified. The specific, testable details of each requirement will be defined in the SRS, where each requirement will also be incorporated to maintain traceability between the two documents.

SR-1: Device Heat Limit

The game shall not cause the device's temperature to exceed a defined safe temperature threshold during normal or extended gameplay. The game shall avoid unnecessary continuous processing and shall stop resource-intensive activities when they are no longer required, so that heat generation stays below this threshold.

Rationale: Excessive processing can raise device temperature. Young children may hold a device for extended periods, so a device that becomes too hot could cause discomfort or minor injury. Keeping device temperature below a defined threshold reduces this risk.

Associated hazards: H1-1

SRS: NFR18

SR-2: Device Performance

The application shall maintain a frame rate at or above a defined minimum on supported devices during normal and extended gameplay, and shall maintain responsive gameplay and smooth visual transitions. The application shall minimize unnecessary processing, rendering, and animation, and shall be tested for performance against this minimum.

Rationale: Slow response times and dropped frames may make the game difficult for young children to interact with and follow. Setting a minimum frame rate and testing against it helps ensure that preparation activities remain usable across supported devices.

Associated hazards: H1-2

SRS: NFR12, NFR13

SR-3: Prevention of Unintentional Gameplay Actions

The game shall prevent repeated touch inputs from triggering multiple navigation actions during a single screen transition. Essential procedure information shall not be skipped as a result of repeated or accidental touch inputs.

Rationale: Young children may repeatedly tap the screen while interacting with the game. If multiple inputs are processed unintentionally, the child may skip important explanations about an ophthalmology procedure. Preventing accidental navigation helps ensure children receive the necessary preparation information.

Associated hazards: H2-1

SRS: FR19; NFR12

SR-4: Gameplay Recovery

The application shall provide a valid recovery path from every gameplay state and handle unexpected or invalid inputs without preventing the child from continuing the activity. The child shall be able to safely restart the current activity without losing previously saved progress.

Rationale: Unexpected inputs or incomplete actions may leave the game in an unusable state. Providing recovery options allows the child to continue preparing for the appointment without

unnecessary frustration or loss of progress.

Associated hazards: H2-2

SRS: FR20, FR22, FR23, FR24; NFR4, NFR17

SR-5: Clinical Approval of Procedure Content

Every procedure shown in the game, including its steps, equipment and the sensations the child may feel, shall be approved by a paediatric eye-care clinician before the game is released. The approval shall be recorded.

Rationale: No team member has clinical training. A procedure shown wrongly means the child's expectations do not match the clinic, which causes the anxiety the game aims to prevent.

Associated hazards: H2-5

SRS: NFR22, NFR27

SR-6: Variation Between Clinics

The game shall tell the child, in words and pictures suited to their age, that their doctor may do things a little differently from the game.

Rationale: Procedures vary between patients, clinicians and clinics (Assumption 5). Telling the child this in advance means a difference at the visit is less likely to surprise or upset them.

Associated hazards: H2-5

SRS: FR9

SR-7: Release of Approved Content Only

The released game shall contain only the version of the educational content that was approved by the clinical and child-life reviewers. Any change to approved content shall be reviewed again before release.

Rationale: An approval is only meaningful if the content the child sees is the content that was reviewed. Late edits, such as wording changes or new art, could reintroduce errors.

Associated hazards: H2-5

SRS: NFR22, NFR27

SR-8: Honest Description of Sensations

For every part of a procedure that the child may feel (such as cold, stinging, a bright light, pressure or blurry vision), the game shall describe the sensation before showing that part of the procedure. The description shall not be skippable, and the game shall not state or imply that a procedure will not hurt or be uncomfortable.

Rationale: Clinicians identified stinging eye drops as a major source of anxiety. A child who has been told about a sensation is not surprised by it, and does not learn to distrust the people who prepared them.

Associated hazards: H2-6

SRS: FR8; NFR2

SR-9: Child-Life Review of Content

All images, sounds, language and story in the game shall be reviewed by a child-life specialist or child development expert before release, confirming that they are suitable for children aged three to seven, contain no frightening material, and make no promise that a procedure will not be uncomfortable.

Rationale: Child-life experts judge emotional suitability, which the clinical review does not cover. This matters more in a game, where story and excitement are added to the medical content.

Associated hazards: H2-6, H2-7

SRS: NFR2, NFR8

SR-10: Medical Disclaimer and Caregiver Acknowledgment

The game shall inform caregivers that the application is intended solely as an educational preparation aid and does not replace professional medical advice, diagnosis, or treatment. The game shall require caregiver acknowledgment of this notice before first use and make the notice accessible from the caregiver procedure guide.

Rationale: The game introduces children to ophthalmology procedures through interactive simulations. Without a clear explanation of its limitations, caregivers may mistake the educational content for professional medical guidance. Requiring acknowledgment communicates the application's intended purpose before use.

Associated hazards: H2-8

SRS: FR1, FR2; NFR32

SR-11: All Preparation Levels Must Remain Accessible

The game shall allow the child to access every ophthalmology preparation level regardless of whether previous levels have been completed. Journey points, completion status, and rewards shall not restrict access to preparation games.

Rationale: The procedures required during an ophthalmology appointment may change unexpectedly. Restricting access to later levels could prevent children from preparing for an examination they may need. Keeping all levels accessible allows children to explore the procedures in any order.

Associated hazards: H2-11

SRS: FR4

SR-12: Encouraging Feedback Without Penalties

The game shall allow children to skip individual exercises without losing previously earned points, coins, or cosmetic rewards. Incorrect responses and skipped exercises shall not trigger negative feedback or prevent the child from completing the preparation level.

Rationale: The application aims to reduce anxiety associated with ophthalmology examinations. Penalizing children for incorrect answers or skipped exercises may cause frustration or discourage participation. Non-punitive feedback allows children to explore procedures at their own pace without pressure to perform correctly.

Associated hazards: H2-12

SRS: FR13, FR18, FR28

SR-13: Termination of Audio and Visual Effects on Exit

When the child exits a preparation level, the game shall immediately stop all audio playback,

animations, and visual effects associated with that level. No level-specific audio or visual activity shall continue after the child returns to the journey map.

Rationale: Children may choose to exit a preparation activity at any time. If audio or visual effects continue after exiting, the child may become confused or distressed by content from an activity they are no longer participating in. Stopping these components ensures that exiting the level fully ends the activity.

Associated hazards: H2-13

SRS: FR21; NFR4

SR-14: Safe Audio and Visual Effects

The game shall avoid unnecessarily loud audio, rapidly flashing effects, and other audio or visual effects that could cause discomfort or distress. Audio shall be adjustable or muted without preventing access to essential educational information.

Rationale: The intended users are children aged three to seven and may be sensitive to sudden or intense audio and visual effects. Keeping effects comfortable reduces avoidable distress while preserving the educational purpose of the game.

Associated hazards: H2-16

SRS: FR15, FR41; NFR15, NFR16

SR-15: Assuring Game Characters and Cosmetic Rewards are Diverse and Inclusive

The player character library shall offer multiple dimensions of diversity (e.g. skin tones, hair textures and styles, varied body types, accessibility and medical-device options, gender expressions etc.) and cosmetic rewards shall include items from a range of cultures and religions.

Rationale: Diverse and inclusive representation lets the child see themselves and their world understanding to support their sense of belonging that carries into the clinic making it a more comfortable environment.

Associated hazards: H2-9

SRS: FR32; NFR31

SR-16: Revealing Correct Answers for Learning Benefit

After an incorrect selection, the game shall indicate all the correct answers within 1 second by at least two non-color cues (e.g., outline, icon, motion) and narration, for the child to select any or all before proceeding with encouraging feedback.

Rationale: Prevents the child from leaving a question unsure of the answer, which reduces anxiety and reinforces learning in a format that supports learning for color-blind children and pre-readers.

Associated hazards: H2-10

SRS: FR16; NFR9, NFR10, NFR12

SR-17: Ensure Quiz Questions have at least One Correct Answer

Every quiz question shall have at least one validated correct answer, checked at build time with questions whose answer keys are invalidated skipped and logged rather than shown.

Rationale: Addresses the data-defect cause and removes quiz questions that don't match so the child never sees a question that cannot be answered correctly.

Associated hazards: H2-10

SRS: FR17

SR-18: Guidance to fix Muted Narration

The game shall switch to visual icon guidance if narration cannot be delivered and show a non-blocking, parent-directed prompt to fix it. The game shall pause and resume at the same point only if audio or visual interruptions (calls, notifications) occur.

Rationale: When narration issues surfaces, the problem is shared to the parent without blocking children who cannot use audio (deaf children). By pausing for spontaneous interruptions it ensures children do not miss content during the game.

Associated hazards: H2-14

SRS: FR15, FR25; NFR9

SR-19: Visual Cues Inclusion during Gameplay

Visual gameplay shall be active in all gameplay states and shall not be disactivatable by the child with any setting to change the visual game instructions parent-gated.

Rationale: Removes the cause where the child turns cues off intentionally or accidentally, while still allowing a parent to adjust for accessibility needs.

Associated hazards: H2-15

SRS: FR14, FR38, FR41; NFR9

SR-20: Reliable Save Data Handling

The game shall validate saved progress before using it and shall handle interrupted or invalid save operations without crashing. A failed save operation shall not overwrite valid previously saved progress with invalid data.

Rationale: Corrupted save data could cause the child to lose completed preparation activities and rewards. Protecting valid saved progress reduces the frustration and loss of functionality caused by unexpected application or device failures.

Associated hazards: H2-18

SRS: FR26, FR27; NFR17

SR-21: Minimization and Protection of Game Data

The game shall store only information required for offline gameplay and shall not store names, contact information, or other unnecessary personal information. Saved game data shall be stored using the platform's application storage mechanisms and shall not be intentionally exposed to other applications. The game, including all packaged third-party libraries, shall not transmit data or initiate network communication during use.

Rationale: The game does not require personal information to provide its core functionality. Minimizing stored information reduces the potential impact of a privacy or security failure, particularly because the application is intended for young children. Keeping saved files private does not prevent a library from sending usage data off the device, so the restriction also applies to network activity.

Associated hazards: H2-19, H2-22

SRS: NFR26, NFR28, NFR29

SR-22: Third-Party Software Validation

Third-party libraries and engine components used by the released game shall be tested as part of the release build. Dependency versions shall be explicitly managed, and changes to third-party software shall be tested before being included in a release.

Rationale: The game depends on third-party software that the development team does not control. A defect, incompatibility, or unexpected dependency update could cause crashes or incorrect behaviour in critical preparation activities.

Associated hazards: H2-20, H2-22

SRS: NFR17, NFR25, NFR28, NFR30

SR-23: Third-Party Asset Review

All third-party art, fonts, sounds, and other assets included in the game shall be reviewed before release for age appropriateness, accuracy, and consistency with the clinician and child-life approved content. A third-party asset shall not be included in the release if it has not completed the required review.

Rationale: Third-party assets are part of the user's experience even though they are not created by the team. An inappropriate or misleading asset could introduce a hazard that would otherwise have been removed from the application's first-party content.

Associated hazards: H2-21

SRS: NFR2, NFR33

SR-24: Accessible, Age-Appropriate Interaction

The game shall present every preparation instruction through short, familiar narration and a visual demonstration, without requiring the child to read more than a few words. Controls shall be large enough and interactions forgiving enough for children aged 3–7 to use.

Rationale: A child can be unable to use a game that runs correctly. Small controls or instructions they cannot follow can stop preparation and leave the child frustrated or anxious. Instructions and interactions need to be tested with the intended ages and abilities, not just checked for game errors.

Associated hazards: H2-3

SRS: FR7, FR12, FR41; NFR3, NFR5, NFR8, NFR9, NFR10

SR-25: Consistent Narration and Animation

During a preparation activity, narration and its visual demonstration shall describe the same step and remain synchronized. After an interruption, both shall resume from the same point.

Rationale: An accurate script and an accurate animation can still teach the wrong thing if they are played together at the wrong time. The child needs to hear the explanation that matches what they are seeing, including after an interruption.

Associated hazards: H2-17

SRS: FR25; NFR14

SR-26: Offline Availability of Preparation Content

After installation, every level and all assets needed to complete it shall be available without an

internet connection. Starting a level for the first time shall not require a download or a response from an external service. The release build shall be tested offline from a fresh installation.

Rationale: A family may open a level in a waiting room with no internet connection. Testing a fresh installation checks that all required content is actually included.

Associated hazards: H2-4

SRS: NFR19, NFR20

7 Roadmap

Table 3 outlines when each SR will be implemented and verified. Proof of Concept (PoC) entries describe small experiments used to investigate safety and feasibility risks. Safeguards apply to the activities being demonstrated, including privacy and offline operation throughout the experiments. Final Delivery entries describe what will be completed and verified by the end of Capstone, with implementation and testing taking place throughout development.

Testing will use the devices and measurement tools available to the team. Final Delivery may use a development build. Actual publication depends on time and app marketplace approval.

Requirements marked "Capstone and Future" will be implemented for the Capstone version and maintained as content and dependencies change. Future work includes expanding content and repeating reviews/testing. It does not mean that the implementation of these requirements is postponed.

The review required by SR-9 is targeted for completion during Capstone, but it depends mainly on reviewer availability. If it cannot be done by the Final Delivery, it will be recorded as outstanding in the final documentation, and final release will remain dependent on that review.

Table 3: Safety Requirements Roadmap

Safety requirements	**Capstone / Future**	**Planned deliverable**	**Rationale**
SR-1 Device heat limit	Capstone	**Final Delivery:** Measure temperature during sustained gameplay on available devices where measurement is possible, and compare it with the SRS limit.	Sustained testing during development helps identify intensive processing. A short demonstration will not show how warm the device gets over time.
SR-2 Device performance	Capstone	**PoC:** Measure response and frame rate on available devices. **Final Delivery:** Verify the SRS performance targets across the completed game.	Early performance measurements help to choose an approach that may work well on the devices families have. Final testing checks the effect of all the content.
SR-3 Unintentional actions	Capstone	**PoC:** Test repeated taps in interactions, preventing any narration from being skipped. **Final Delivery:** Apply input protection to all transitions and verify that repeated taps cannot skip essential content.	Sample interactions allow investigation of accidental input before the full navigation is built. Applying the same protection across the game ensures that later activities do not let repeated taps skip anything.

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Safety requirements	Capstone / Future	Planned deliverable	Rationale
SR-4 Gameplay recovery	Capstone	**PoC:** Try stopping and restarting interactions where applicable. **Final Delivery:** Complete recovery paths that preserve progress and test unexpected inputs from every gameplay state.	These experiments help identify how a child could get stuck or need to restart. Full recovery can then be implemented and tested alongside progress, rewards, and remaining activities.
SR-5 Clinical approval	Capstone	**PoC:** Seek clinician review of a sample eye drop activity. Record corrections and approval status. **Final Delivery:** Obtain recorded approval for every included procedure.	Reviewing early helps to assess whether accurate content can be produced and reviewed in a reasonable amount of time. Every included procedure still needs recorded approval before release.
SR-6 Variation between clinics	Capstone	**PoC:** Include an age appropriate explanation that the doctor may do things differently. **Final Delivery:** Verify that every level includes this explanation.	This message belongs in the PoC content because it affects what the child expects during the appointment. It is also needed for each new level added.
SR-7 Approved content only	Capstone and Future	**PoC:** Start content version and approval records. **Final Delivery:** Check content in the final Capstone build against the versions approved by reviewers. **Future:** Review changes again before release.	Keeping approval records from the start makes it easier to track what was reviewed. This needs to continue after capstone because new wording, art, or procedures can change content that was approved previously.
SR-8 Honest sensations	Capstone	**PoC:** Include honest sensation descriptions in the sample content. **Final Delivery:** Verify descriptions and their timing in every applicable procedure step.	The eye drop content is central to the PoC, so explaining sensations is part of the initial review. Checking all procedures later ensures that new content provides the same level of detail.
SR-9 Child-life review	Capstone and Future	**PoC:** Arrange initial review of the sample content and record feedback and review status. **Final Delivery:** Target completed review of all final content, with findings addressed before release. **Future:** Review new or changed content before release.	Reviewer availability may affect the timeline, so this needs to be arranged early. The review supports the goal of reducing anxiety and is already required before release. If it is delayed, it still remains a requirement.
SR-10 Medical disclaimer	Capstone	**Final Delivery:** Implement the caregiver notice, require and store acknowledgement, and retain access through the caregiver guide.	The notice is part of the application's setup flow. Testing it with the rest of the game checks that caregivers see it prior to use.

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Safety requirements	Capstone / Future	Planned deliverable	Rationale
SR-11 All levels accessible	Capstone	**PoC:** Where activity selection is demonstrated, allow direct access without requiring completion of another activity. **Final Delivery:** Verify that every level remains accessible regardless of progress or rewards.	Open access applies to any activity selection used in the experiments. The full journey needs to be checked once progress and rewards are added.
SR-12 Encouraging feedback	Capstone	**PoC:** Try encouraging and helpful feedback for incorrect actions in a sample interaction. **Final Delivery:** Complete individual activity skipping and reward protection, and verify that mistakes or skipped exercises do not result in penalties.	Encouragement is already promised in the PoC plan and supports the purpose of reducing anxiety. The wider checks become possible once the game has more features (such as rewards) to test together.
SR-13 Stop effects on exit	Capstone	**PoC:** Stop audio and animation when an experiment is exited. **Final Delivery:** Verify that every level stops all associated effects on exit.	The effects need to stop when the user exits. Each later activity needs to be checked as well since it may introduce different effects.
SR-14 Safe audio and visuals	Capstone	**PoC:** Avoid harsh audio and flashing visuals. **Final Delivery:** Complete audio controls and visual access to essential information, and review all effects.	The sample interactions should already be comfortable to use. Completing the controls before final testing allows checking of muted gameplay, while the final asset review covers sounds and visuals added later on.
SR-15 Inclusive characters and rewards	Capstone and Future	**Final Delivery:** Provide a manageable character and accessory collection that is diverse. **Future:** Expand the range of characters and cosmetics.	A smaller collection can cover the promised representation within the Capstone timeline. More options can be added later.
SR-16 Correct answer reveal	Capstone	**Final Delivery:** Implement correct answer reveal with narration and (non-colour) visual cues. Verify timing, cues, and feedback for all questions.	Showing the answer is part of how a question teaches the child, so it should be implemented with the question system.
SR-17 Valid quiz answers	Capstone and Future	**Final Delivery:** Implement build-time answer validation as questions are added during Capstone, excluding and logging invalid questions in every build. **Future:** Run it for every build as questions change.	The validation needs to exist when questions are first added so that invalid questions cannot reach the child. Keeping the check in the build process also covers future content.

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Safety requirements	Capstone / Future	Planned deliverable	Rationale
SR-18 Muted narration and interruptions	Capstone	**Final Delivery:** Implement visual guidance, a caregiver prompt, and handling for pausing/resuming. Test muted audio and interruptions on team devices.	This need to work for all activities. Testing it on team devices helps to find platform differences and check that children do not miss any instructions.
SR-19 Visual guidance	Capstone	**PoC:** Try visual guidance in the sample interactions. **Final Delivery:** Complete caregiver settings and verify that guidance remains available in every gameplay state.	Visual guidance is a core part of making the game usable for children who cannot read.
SR-20 Reliable saved progress	Capstone	**Final Delivery:** Implement save validation, protection of progress, and recovery. Test failed saves and subsequent application launches.	Save handling needs to be tested with progress and rewards, rather than added at the end. Testing it during development allows testing of failure cases that may not appear during normal demonstrations.
SR-21 Data protection and no networking	Capstone and Future	**PoC:** Keep any prototype data minimal and local, with no network communication. **Final Delivery:** Inspect stored data and check for outbound requests (including from libraries). **Future:** Repeat checks after future changes.	These are existing data constraints on how the game is built, so they need to guide development right away.
SR-22 Third-party software	Capstone and Future	**Final Delivery:** Test dependencies in the final Capstone build and check for known vulnerabilities. **Future:** Test dependency changes before release.	Fixed versions make builds consistent, while tests and vulnerability checks help find problems introduced by libraries. These checks need to continue during and after Capstone whenever dependencies change.
SR-23 Third-party assets	Capstone and Future	**PoC:** Review the assets used in the sample interactions. **Final Delivery:** Complete review of every included asset. **Future:** Review additions and replacements before release.	External assets affect what the child sees and hears. Reviewing them as they are introduced avoids unsuitable material making its way into the game.
SR-24 Accessible interaction	Capstone and Future	**PoC:** Try one narrated, visually guided interaction with appropriately sized controls. Record completion time and assistance. **Final Delivery:** Complete available usability and accessibility checks. **Future:** Reassess changed content and interactions.	Early feedback helps shape the activities, but adult testing alone cannot establish usability for children or a reduction in anxiety. Child evaluation depends on consent and ethics approval, so any missing evidence needs to remain clear.

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Safety requirements	Capstone / Future	Planned deliverable	Rationale
SR-25 Narration and animation	Capstone and Future	**PoC:** Measure synchronization in sample interactions. **Final Delivery:** Verify every procedure step, including after interruptions. **Future:** Repeat checks when changes are made to audio, visual, etc.	Synchronization is a part of the PoC plan. Checking it again with final content and after changes helps ensure the child always hears the explanation that matches the step shown.
SR-26 Offline preparation	Capstone and Future	**PoC:** Run the sample interactions and media in an offline mode. **Final Delivery:** Test every level offline from a fresh development installation of the final Capstone build on available devices. **Future:** Repeat the check for releases with new/changed content.	Offline operation is a key part of the project's scope. Testing with a fresh install checks that all assets are packaged with the game, including content added after the PoC.

8 Appendix — Generative AI Use Disclosure

8.1 AI Tools Used

- ChatGPT (GPT-5.6 Luna, GPT-6, GPT-6.1 Sol)
- Claude (Claude Opus 5.5, Medium Effort)

8.2 How and Where Used

ChatGPT and Claude were used as supporting tools during the development and formatting of this document. Its primary purpose was to assist with LaTeX formatting, improve wording and clarity, and brainstorm potential ideas relevant to the hazard analysis.

The following sections were assisted by AI:

- In Section 1 (Introduction) and Section 2 (Scope), Opus 5.5 was used to generate an initial draft based on our problem statement document. The team then manually refined the content.
- In Section 5.2 (FMEA Table), Claude was used to help with the table formatting and the creation of a numbering system. This allowed us to quickly link the hazards to the security requirements without worrying about the order of which items appear first.
- In Section 4 (Critical Assumptions), ChatGPT was used to brainstorm potential assumptions regarding the application's operating environment, intended users, and external dependencies. These suggestions were reviewed and refined to ensure they were relevant to the project's scope.
- In Section 6 (Safety and Security Requirements), ChatGPT was used to ensure that additional safety and security requirements that were brainstormed didn't already overlap with existing SRS NFRs. ChatGPT was also used to refine the requirement wording and help make sure the rationale was clear for SR-24, SR-25, and SR-26. In addition, ChatGPT was used to assist in linking FRs and NFRs from the SRS to their associated SRs in this section. Claude was used to identify clear and concise wording for requirements descriptions.
- In Section 5 (FMEA) and Section 6 (Safety and Security Requirements), ChatGPT and Opus 5.5 were used for brainstorming and idea generation, which the team then used to aid us in coming up with hazards and associated requirements. ChatGPT was also used to help refine the wording of H2-3, H2-4, H2-17, and H2-22, and assist in ensuring that their addition was not redundant given existing hazards. It was used to help identify if new safety requirements were needed (or if the existing requirements provided sufficient coverage) as well.
- In Section 7, ChatGPT was used to set up the format of the roadmap table (including links to all SRs), to help improve the wording of the introductory paragraphs as well as table entries, and enhance the conciseness of the rationale for each SR's placement in the roadmap. It was also used to compare the roadmap to Appendix B in the SRS, to ensure alignment in the timelines and deliverables between the two documents. Its suggestions for improving consistency were implemented by the team.
- For document formatting, ChatGPT was used to assist with converting and formatting written content into appropriate LaTeX structures, including lists, tables, and section organization.

8.3 Verification of AI-Generated Content

All AI-assisted content was reviewed before being incorporated into the document. The team checked that the suggested assumptions, hazards, and safety requirements were relevant to the project and consistent with the existing Problem Statement and Goals, Development Plan, and intended application functionality identified in SRS document. AI-generated suggestions were evaluated against the project's actual scope and the hazard analysis methodology presented in the course lectures. Suggestions that were inappropriate, unnecessary, or inconsistent with the team's decisions were removed or revised. For LaTeX formatting,

the team verified that the document compiled successfully and that the resulting output was formatted as intended.

The team remains responsible for the accuracy, completeness, consistency, and appropriateness of the final document, regardless of whether content was generated or modified with AI assistance.

9 Appendix — Reflection

Questions 1 and 2 were answered individually, while Questions 3 and 4 were answered as a team. The team responses are provided under Team Reflection, while the individual responses are provided under each respective team member's name.

Ethan McMehen

1. **What went well while writing this deliverable?**

   I found what went well in writing this deliverable was identifying hazards and producing the associated safety and security requirements. At the stage when I began working, my teammate Jake wrote the System Components and Boundaries documentation. This clarified what our components for the application would be, making it easier to identify hazards that could arise in each of these components.

   I wrote hazards that stemmed from Quizzes, Save Data, and Educational Content which I was able to identify relatively easily because of the descriptions by Jake. Translating the hazard into a failure, effect, cause, mitigation also went well because that is a simplified pattern for me to think through the hazard and its solution. Then using the recommended solutions, I could dive into more detail about a safety requirement to generate to support the mitigation strategy and risk minimization of the hazard.

   From this whole process, I learned how using FMEA is a great method to identify and document hazards. FMEA is also a strong tool to develop safety requirements directly connected to the scope and objectives of the project.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   The two pain points I experienced was avoiding repetitiveness in my hazard and requirements generation. Since a couple of team members worked on the FMEA table and requirements before I did, I had to check their hazards and requirements in depth to avoid generating similar risks, hazards or requirements which took me about an hour to fully understand and comprehend their work. Only then was I able to work on my own elicitation. Additionally, the goal of safety and security requirements is to generate new requirements separate from the NFRs in the SRS to ensure these requirements directly connect to the hazards. From this, I also had to double check my new requirements with the NFRs to avoid repeating myself. Altogether, this took a lot of time and extreme attention to detail.

   To solve these pain points, I wrote my own cross-reference notes. I identified the system components that the previous hazard and requirements written targeted. Then I used the remaining system components as my main set to write hazards and requirements for. This enabled me to avoid repeating my teammates while still finding original and positive work to contribute to the HA document. Regarding NFRs, I did just read through the NFRs side by side with my new requirements to ensure they were different.

   From these pain points I learned that cross-referencing can be done well if I identify the overall classes of work previously done before me. That way I can focus on the potential holes that exist within the documentation. As a result, it means I perform meaningful work for our deliverable and I contribute overall to the team.

Jake Finlay

1. **What went well while writing this deliverable?**

   What went best was that we split up the work before we started writing. Every team member had their own assigned issues on GitHub. This made it easy to find what to work on next. It also made it clear who was responsible for each part of the document.

   Other than the pain points mentioned below, everything else went fine. Nothing else noteworthy to write here.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   One pain point I experienced during this deliverable was that some sections depend on others. Some of my teammates were waiting on sections I was working on. To keep the team moving, I had to finish those sections first. This meant putting off other work that I had planned to complete.

   It was also hard to write content that my teammates had not already covered. I had to keep checking their pull requests and the PDFs produced by our CI pipeline. This was the only way to make sure I was not repeating something from another section. To solve this, I downloaded everyone's PDFs and kept them open on my second monitor. I read through them before I started writing. After that, writing my sections was fairly straightforward.

   The last pain point was linking the safety and security requirements to the hazards in the FMEA table. The order of the requirements and hazards was likely to change as everyone's work was merged. This would cause them to be mislabelled, or force a teammate to renumber all of their requirements and hazards by hand. To solve this, I created a unique identifier system. Each requirement and hazard is defined with its own identifier. Every reference links to that identifier instead of a number. If the order changes, the numbers and links update automatically. This made it much easier to link everything together and get the traceability I wanted. I plan to use the same system in the SRS when I start working on it.

Ahsan Muzammil

1. **What went well while writing this deliverable?**

   One aspect that went well was how the team organized and divided the work. By creating and assigning GitHub issues early, everyone had a clear understanding of their responsibilities and what needed to be completed. This also allowed us to track each team member's progress and check that the workload was distributed fairly. Having clearly assigned tasks helped us avoid confusion and made it easier to coordinate our contributions throughout the deliverable.

   Another positive aspect was the review process. Team members reviewed each other's contributions within a reasonable amount of time, which helped us identify mistakes, provide feedback, and merge changes without significant delays. This was particularly helpful because the Hazard Analysis required consistency between the identified hazards, their causes, recommended actions, and the corresponding safety and security requirements.

   Writing the Critical Assumptions section also went well because it did not depend on other team members completing their sections first. This allowed me to work independently and complete my assigned section without delays. I was able to identify and document the assumptions relevant to the system while the rest of the team continued working on other sections.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   One challenge was contributing to Section 5 (Failure Mode and Effects Analysis) and Section 6 (Safety and Security Requirements). These sections depended on information from earlier sections, particularly the System Boundaries and Components section. Since those sections were still being developed, I could not begin my contributions as early as I had planned.

   To resolve this, I communicated with the teammates responsible for the dependent sections and asked them to prioritize completing their work so I could proceed with mine. In the meantime, I focused on completing the Critical Assumptions section. This allowed me to continue making progress rather than waiting for the other sections to be finished.

   Another challenge was having to reconsider some of the hazards after receiving feedback from our supervisor. For example, the supervisor recommended that all preparation games remain accessible rather than requiring children to unlock levels through progression. This highlighted a risk where

children might be unable to prepare for a procedure because they had not completed earlier levels. As a result, I identified the restricted level access hazard (H2-11) and added a corresponding safety requirement to ensure that all preparation levels remain accessible regardless of the child's progress.

This required revisiting the application's design and ensuring that the FMEA and safety requirements reflected the supervisor's feedback. Although this took additional time, it helped me understand how stakeholder feedback can reveal risks that were not initially considered.

Mohammad Bilal

1. **What went well while writing this deliverable?**

   One thing that went well was writing the Introduction and Scope sections. Having already spent significant time working on this project, I had a good understanding of its purpose and the risks it was meant to address. This made it easier to explain the motivation behind the hazard analysis and identify what the document needed to cover. The team was also on the same page about the project's goals and scope, so there was little disagreement about what to include.

   Another thing that went well was the division of tasks and use of GitHub. Learning from some of the issues we experienced during the first deliverable, the Problem Statement and Development Plan, we split the work earlier and made sure everyone knew what they were responsible for. This gave us a clearer idea of what needed to be completed and helped us avoid some of the coordination issues we had experienced before.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   One pain point was trying to work on the document in parallel. Although splitting tasks allowed everyone to work independently, many sections depended on information from other sections. For example, the hazards in the FMEA had to be based on the system components, while the safety requirements had to address the hazards identified in the FMEA. This made it difficult to complete some sections without knowing what had already been decided in the earlier sections. As a result, working on multiple sections at the same time sometimes meant that someone had to wait for another person's work before they could continue.

   Another challenge was making sure that different sections did not contain conflicting or repeated information. Since multiple people were writing about related parts of the system, it was possible for the same idea to be explained more than once or for two sections to make slightly different assumptions. This was especially important for the hazard analysis, since the hazards, recommended actions, and safety requirements needed to agree with one another. We had to keep checking the other sections and reviewing the merged document to make sure everything was consistent.

   We resolved these issues by having team members work more sequentially when their sections depended on one another. Instead of everyone continuing to work on related sections at the same time, we prioritized completing and merging the earlier sections before moving on to the dependent ones. This gave each person a more stable version of the document to work from and reduced the chances of conflicting or duplicated content. Although this approach sometimes meant waiting for another section to be completed, it made the overall process easier to manage and helped us maintain consistency throughout the deliverable.

Jackson Beach

1. **What went well while writing this deliverable?**

   What went well was that the main section I was responsible for, Section 7 (Roadmap), built on the earlier sections completed by the team. Everyone came up with clear and actionable safety requirements based on the hazards they identified. This made it relatively straightforward to figure out when each requirement could be feasibly met across the project's deliverables. We also split up the larger Sections

5 and 6 well, so everyone could contribute their perspective without being overwhelmed by taking on an entire section.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   The main pain point I found was keeping the SRS and HA consistent. Each new safety requirement needed to be linked to and covered by FRs and NFRs in the SRS. There were a sizeable number of safety requirements identified from the hazards, as well as a sizeable number of FRs and NFRs present in the SRS. Keeping track of these links, in addition to making sure the links were to relevant requirements, became tricky. As the document grew, it also became harder to spot redundant requirements and hazards. We addressed these issues through the following strategies: frequent discussions on Discord, giving and receiving thorough peer reviews on GitHub, and using AI tools where needed to help find and fill gaps in the links between the SRS and HA.

Team Reflection

3. **Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?**

   Before completing the Hazard Analysis, the team had already considered several risks related to the application's purpose and target users. These included inaccurate representations of ophthalmology procedures (H2-5), frightening educational content (H2-7), and interactions that young children may find difficult to understand or use (H2-3). The team had also considered risks involving the privacy of locally stored information (H2-19) and the unavailability of preparation activities without an internet connection (H2-4). These concerns had influenced the application's requirements before the Hazard Analysis began.

   During the Failure Mode and Effects Analysis (FMEA), the team identified additional hazards by examining how individual system components could fail and how those failures might affect children or caregivers. Table 4 identifies the hazards considered by each team member during the analysis and explains how they were identified.

Table 4: Hazards Identified During the FMEA

Team Member	Hazards Identified	How the Hazards Were Identified
Ahsan Muzammil	H2-12: Penalties for incorrect answers or skipped exercises. H2-1: Repeated touch inputs skipping important information. H2-8: Educational content mistaken for medical guidance. H2-11: Preparation levels becoming inaccessible. H2-13: Audio and visual effects continuing after exiting a level.	These hazards were identified by reviewing the gameplay and journey based use cases (from SRS document) alongside the system components. Considering how children might interact with the games highlighted the possibility of accidental repeated inputs and negative feedback causing frustration. Reviewing the educational purpose of the application highlighted the risk of caregivers mistaking the content for medical advice. The supervisor's recommendation that all preparation games remain accessible helped identify the risk of restricted level access. Finally, reviewing the use case (from SRS document) for exiting a preparation level highlighted the possibility of audio and animations continuing after the child had left the activity.

Team Member	Hazards Identified	How the Hazards Were Identified
Jake Finlay	H2-5: Procedures shown with the wrong steps, order or equipment. H2-6: Unpleasant sensations left out or played down. H2-7: Frightening images, story moments or language.	These hazards came from the clinical accuracy and age-appropriateness risk in the Development Plan. That risk already covered wrong steps and missing sensations. When the Educational Content component was analysed in the FMEA, the single risk was split into separate failure modes. Each one has its own causes and mitigations. Inaccurate procedures and frightening content had been considered before. Hiding unpleasant sensations was the new hazard. It came from asking how content could be accurate and still harm the child. Writers may try to reassure the child by saying a procedure will not hurt. The child then feels betrayed when the eye drops sting. This led to the requirement that every sensation is described honestly and cannot be skipped.
Ethan McMehen	H2-9: Diversity and inclusion for characters H2-10: Failure to show correct answer(s) for learning benefit H2-14: Narration not delivered during gameplay H2-15: Visual gameplay descriptions is not shared to children	These hazards were primarily identified by reviewing stakeholder discussions and characteristics. Supervisors asked to allow for character selection and customization for the children to feel comfortable about the environment. Supervisors and clinicians in particular seek that children will learn from the game to prepare for the examinations. Failure to provide answers to questions in quizzes could increase anxiety from misunderstanding of examination expectations. When reviewing accessibility, the narration and visual gameplay hazards are present to acknowledge the possibility of visually impaired children learning at a disadvantage in the event a narration or visual gameplay user error occurs.

Team Member	Hazards Identified	How the Hazards Were Identified
Mohammad Bilal	H1-1: Device becoming excessively hot during use. H1-2: Slow response times or dropped frames during gameplay. H2-2: Gameplay becoming stuck after an unexpected or invalid action. H2-16: Sudden loud audio or rapidly flashing visual effects. H2-18: Saved progress becoming corrupted or reset. H2-19: Unnecessary personal information in save data, or save data accessible to other applications. H2-20: A third-party library or engine component crashing or behaving incorrectly. H2-21: Third-party assets with inappropriate or misleading content.	These hazards were identified by going through the system components listed in the Components section one at a time and asking how each one could fail and how the failure would affect the child. For the Device, the limits of a phone or tablet led to the risks of overheating during long play and of poor performance. For Gameplay, considering unexpected or invalid actions from a young child led to the risk of the game becoming stuck with no way to continue. For Audio and Visuals, the child's sensory comfort led to the risk of sudden loud sound and flashing effects. For Save Data, the progress and settings stored on the device led to the risks of lost progress and of storing unnecessary personal information. For Third-party Software and Assets, the use of code and art that the team did not write led to the risks of a library failing and of an asset containing content that is not suitable for children.
Jackson Beach	H2-3: Instructions or interactions that children cannot understand or use. H2-4: Preparation levels or required content being unavailable offline. H2-17: Narration describing a different procedure step from the animation being shown. H2-22: Included libraries sending data off the device.	The first hazard was identified as the SRS contains FRs and NFRs that focus on ensuring an age-appropriate experience for paediatric patients. Existing SRs partially covered these, but collectively they did not specify the exact risk of a child not being able to understand instructions and how their frustration levels could impact what they gain from the game. The second hazard was identified as the system constraints in the SRS require offline operation of the application, and no hazard at the time covered this. The third hazard was identified as the SRS has FRs and NFRs related to timing alignment for the content, but no hazard highlighted the risk of audio-visual inconsistency and how that could impact a child's learning about the appointment. The fourth hazard also came from the system constraints in the SRS, as transmitting data from the device over a network is prohibited. What was missing from the table at the time was an analysis of the risk posed by third-party dependencies included in the game collecting and transmitting data.

The FMEA helped the team move beyond identifying general risks and examine specific failures that could occur during gameplay. By considering the causes and effects of each failure, the team was able to develop additional safety and security requirements to reduce these risks.

4. **Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?**

   Two other types of risk in software products are privacy risks and financial risks. Neither involves physical harm, but both can seriously affect the people who use a product and the organizations that build it.

   Privacy risks come from the personal data that software collects. Many products store names, locations, contact information and payment details. If this data is exposed through a breach or a design flaw, users can face identity theft, fraud, stalking or embarrassment. The harm falls on the users, yet they usually have no control over how their data is stored or protected. A breach can also lead to legal penalties for the organization and a lasting loss of trust in the product. These risks are especially important for our project, because our users are young children. Children cannot judge what information is safe to share, and their data is protected by stricter laws and app store rules. This is one reason our game is designed to work offline and to collect no personal information at all.

   Financial risks come from software failing to do its job correctly or failing to stay available. A bug in a payment system or a financial calculation can cost users or organizations money directly. Downtime can also bring a business to a halt, since many organizations rely on software for their daily operations. Because a single piece of software is often used by thousands of people at once, one failure can lead to large and widespread costs. Financial risk can also come from the design of a product itself. For example, games with in-app purchases can lead children to spend money without understanding it. Considering these risks early allows a team to design them out, rather than repairing the damage after the product is released.

References

[1] Nancy G. Leveson. *Engineering a Safer World: Systems Thinking Applied to Safety*. Engineering Systems. MIT Press, Cambridge, MA, 2011.